

Math Bootcamp: Content to be Covered

For Incoming MSc Economics Students

Math Camp 2026

Overview

- **Objective:** Bridge the gap between a weaker undergraduate math background and MSc-level economics courses.
- **Duration:** 3 hours per day for 5 days.
- **Method:** Focus on intuition, fundamentals, and confidence-building. Rigorous proofs and advanced techniques are deferred to fall courses; only simple proofs are included where necessary.
- **Emphasis:** Conceptual topics to be covered throughout graduate microeconomics, macroeconomics, econometrics, and mathematical economics.

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1 Day 1 — Foundations (Sections 1.1–1.9)

Approximate duration: 3 hours. See Day 1 slides (Sections 1.1–1.9).

1.1 1.1 — Notation and Number Systems

- Number systems: $\mathbb{N}$ (counting), $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{R}^n$; nesting $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$.
- Symbols: $\in$, $\subset$, $\forall$, $\exists$, $\Rightarrow$, $\Leftrightarrow$.
- Scalars versus vectors: $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$; dot product $\mathbf{p} \cdot \mathbf{x} = \sum_i p_i x_i$.
- Practice: identify the smallest set containing a given object (e.g. counts in $\mathbb{N}$; prices in $\mathbb{Q}$; order vectors in product sets).

1.2 1.2 — Logic and Mathematical Statements

- P **only if** Q ($P \Rightarrow Q$): Q necessary for P .
- P **if** Q ($Q \Rightarrow P$): Q sufficient for P .
- P **iff** Q ($P \Leftrightarrow Q$): necessary and sufficient.
- Contrapositive: $\neg Q \Rightarrow \neg P$ equivalent to $P \Rightarrow Q$.
- Running example: IKEA budget set $\mathcal{B} = \{\mathbf{x} \in \mathbb{N}^3 : \mathbf{p} \cdot \mathbf{x} \leq B\}$; practice with $\in$, $\subset$, $\forall$, $\exists$, $\Rightarrow$, and contrapositive.

1.3 1.3 — Sets and Set Operations

- Notation: $\in$, $\notin$, $\subset$, $=$, $\emptyset$, roster and set-builder forms.
- Union, intersection, complement, difference; Cartesian product $A \times B$.
- Economic sets: budget sets, consumption sets; product spaces such as $\mathbb{Q} \times \mathbb{N}$ versus constraint sets like $\mathcal{B}$.
- Practice: describe $\mathcal{B} \cap \mathcal{H}$, $\mathcal{B} \cup \mathcal{H}$, $\mathcal{B} \setminus \mathcal{H}$; distinguish elements from subsets.

1.4 1.4 — Functions

- Function $f: A \rightarrow B$: domain, codomain, range $f(A) \subseteq B$.
- Injective (one-to-one) and surjective (onto); codomain versus range.
- Operations: $(f \pm g)$, (cf) , (fg) , composition $(f \circ g)(x) = f(g(x))$.
- A rule must assign **exactly one** output per input (not a correspondence).
- Practice: domain, codomain, range, injectivity, and surjectivity (e.g. partner age as a function of your age).

1.5 1.5 — Supremum and Infimum

- Example first: reservation price / willingness to pay when the purchase set $P = \{p \in \mathbb{R}_+ : p < \bar{p}\}$ has no maximum but $\sup P = \bar{p}$.
- $\max S$ / $\min S$ versus $\sup S$ / $\inf S$ (least upper bound, greatest lower bound).
- When extrema are attained versus only approached (e.g. $S = [0, 1)$, $\sup S = 1 \notin S$).
- If $\max S$ exists, $\max S = \sup S$; always $\inf S \leq x \leq \sup S$ for $x \in S$.

1.6 1.6 — Algebra Revision

- Taylor expansion around a : $f(a+h) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} h^k$ (first-order: $f(a+h) \approx f(a) + f'(a)h$).
- Log and exponential rules; series $\ln(1+x)$ and e^x .
- **Why logs:** $\Delta \ln x \approx \Delta x/x$ tracks percentage changes; elasticity $\varepsilon_{y,x} = \partial \ln y / \partial \ln x$; log-log regression.
- Cobb–Douglas: $U = x_1^\alpha x_2^{1-\alpha}$, $Y = AK^\alpha L^{1-\alpha}$; linear in logs; constant elasticities and expenditure shares.
- Local log approximation of any smooth $u > 0$ near $\bar{\mathbf{x}}$; Cobb–Douglas as a modeling choice when used globally.
- Homogeneity of degree k : $f(t\mathbf{x}) = t^k f(\mathbf{x})$; Cobb–Douglas utility and production are degree 1 (CRS: double inputs $\Rightarrow$ double output).

1.7 1.7 — Single-Variable Derivatives

- Formal definition: $f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$; meaning as slope and marginal rate (MC , MU , MP_K).
- Rules: constants, sums, power, product, quotient, chain rule; $(e^x)' = e^x$, $(\ln x)' = 1/x$, $(a^x)' = a^x \ln a$; inverse-function rule (stated).
- Practice: cubic cost, monopolist profit, Cobb–Douglas $Y(K)$, composite cost (chain rule).
- **Euler’s theorem** (for C^1 homogeneous f of degree k): $\sum_i x_i f_{x_i} = k f(\mathbf{x})$ (proof via $t \mapsto f(t\mathbf{x})$).
- Adding-up problem: under CRS and competitive factor markets, $PQ = rK + wL$; IRS ($k > 1$) and DRS ($k < 1$) and payment exhaustion.

1.8 1.8 — Multivariate Calculus

- Level curves $\mathcal{L}_c = \{(x, y) : f(x, y) = c\}$ and contour maps; 3D slice picture for $z = f(x, y)$.
- Roadmap: partials $\Rightarrow$ total differential $\Rightarrow$ directional derivative $\Rightarrow$ cross-partial (Schwarz).
- Partial derivatives (limit definition; fix one variable); $MP_K = \partial Y / \partial K$ with L fixed.
- Total differential: add/subtract trick, MVT on each bracket, $dz = f_x dx + f_y dy$; Cobb–Douglas output example $dY = MP_K dK + MP_L dL$.
- Gradient ∇f : perpendicular to level curves; points uphill; $D_{\mathbf{u}} f = \nabla f \cdot \mathbf{u} = \|\nabla f\| \cos \theta$.
- Directional derivative along unit direction $\mathbf{u}$; example: subsidy requiring input changes in ratio 2:1.
- Cross-partial and Schwarz’s theorem (stated); Y_{KL} and complements versus substitutes.

1.9 1.9 — Limits

- Motivation: Berkeley’s critique of “infinitely small” h in early calculus.
- Sequence limit (ε – N): quantifier order and convergence.
- Function limit (ε – δ): $0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon$; example $f(x) = x + 1$ for $x \neq 1$, $f(1) = 3$, $\lim_{x \rightarrow 1} f(x) = 2 \neq f(1)$.

- Continuity at a : $\lim_{x \rightarrow a} f(x) = f(a)$; differentiability at a : derivative limit exists.
- Differentiable $\Rightarrow$ continuous; converse fails ($|x|$ at 0; Weierstrass function).
- Practice: ε - δ proofs for $f'(a) = 2a$ when $f(x) = x^2$ and for $\lim_{x \rightarrow 1} f(x) = 2$.

1.10 Appendix (Slides)

- Derivations of single-variable rules: constants, sums, powers, product, quotient.
- Key limits for $(e^x)'$ and $(\ln x)'$; exponential and logarithm derivative proofs.

2 Day 2 — Calculus II and Linear Algebra: Implicit Functions, Integration, and Matrix Methods

Approximate duration: 3 hours.

2.1 Implicit Function Theorem

- Explicit versus implicit definitions of relationships.
- What problem the Implicit Function Theorem solves: local solvability and comparative statics.
- **Core:** statement and use in one variable ($dy/dx = -F_x/F_y$; $dy^*/d\theta = -F_\theta/F_y$).
- Sign of F_y and direction of local response.
- Application: market equilibrium comparative statics (demand shifter and equilibrium price).
- **Further study (not required):** several variables and matrix/Jacobian form.

2.2 Integration

- Riemann integral: area under a curve.
- Fundamental Theorem of Calculus (statement and intuition).
- Definite versus indefinite integrals.
- Integration as accumulation.
- Link to probability: PDF and CDF; CDF as integral of PDF; expected value as integral (conceptual).
- Inverse relationship between differentiation and integration (conceptual).

2.3 Leibniz Rule and Interchanging Differentiation and Integration

- Leibniz product rule (review): derivative of a product of functions.
- **Fixed limits:** differentiation under the integral sign when the integrand depends on a parameter but the bounds do not.
- **Variable limits:** when the upper and lower bounds of integration depend on the choice variable (parameter), the derivative has two extra *endpoint* terms from the moving limits, plus an integral of the partial derivative of the integrand.
- Informal statement: if $F(\theta) = \int_{a(\theta)}^{b(\theta)} f(x, \theta) dx$, then $F'(\theta)$ equals $f(b(\theta), \theta) b'(\theta) - f(a(\theta), \theta) a'(\theta)$ plus $\int_{a(\theta)}^{b(\theta)} \frac{\partial f}{\partial \theta}(x, \theta) dx$ (under regularity conditions).

- Economic motivation: consumer surplus, areas under shifting demand curves, and other objects where both the integrand and the integration region change with a parameter.
- When we may interchange $\frac{d}{d\theta}$ and $\int$: continuity, differentiability, and boundedness (conceptual); dominated convergence as the measure-theoretic idea behind safe interchange (stated, not proved).
- When interchange fails: discontinuities, unbounded integrands, or limits that move irregularly.
- Distinction from Fubini (swap order of integrals), MCT, and LDCT (swap limit and integral); preview on Day 3 via uniform convergence.

2.4 Mean Value Theorem

- Geometric intuition: existence of a tangent parallel to a secant.
- Rolle's theorem as a special case (statement only).
- Economic uses of the Mean Value Theorem: average versus marginal quantities; bounding changes over intervals.

2.5 Taylor Expansion and Linearization

- First-order Taylor approximation.
- Linearization of nonlinear functions around a point.
- Interpretation as local first-order comparative statics.
- When linearization is useful versus when it fails (curvature, nonlinearity).
- Preview of second-order Taylor expansion (conceptual only).

2.6 Vectors and Matrices

- Vectors in $\mathbb{R}^n$: notation, components, and geometric interpretation.
- Matrices: dimensions, entries, and matrix–vector multiplication.
- Basic matrix operations: addition, scalar multiplication, and multiplication.
- Transpose of a matrix or vector.
- Identity matrix and role in matrix algebra.

2.7 Linear Systems, Inverses, and Determinants

- Systems of linear equations in matrix form ($A\mathbf{x} = \mathbf{b}$).
- Inverse of a square matrix: definition and when it exists.
- Determinant: definition for 2×2 and 3×3 matrices; link to invertibility (conceptual).
- Linear independence and dependence of vectors (brief, intuitive treatment).
- Rank of a matrix (conceptual: number of linearly independent rows or columns).

2.8 Jacobians and Economic Applications of Linear Algebra

- Jacobian matrix as a collection of partial derivatives.
- Dot product and its economic interpretation (e.g., expenditure, inner products in econometrics).
- Quadratic forms (brief introduction: notation and positive/negative definiteness, stated not proved).
- How matrix notation simplifies multivariate comparative statics; full IFT in matrix form (further study).

3 Day 3 — Real Analysis: Topology, Convergence, and Existence

Approximate duration: 3 hours.

3.1 Metric Spaces and Basic Topology in $\mathbb{R}$ and $\mathbb{R}^2$

- Points, distances, and neighborhoods.
- Open sets: definition and intuition.
- Closed sets: definition and intuition.
- Interior, boundary, and closure (conceptual).
- Economic relevance: budget sets, feasible sets, production sets; open versus closed constraints.

3.2 Convergence

- Sequences and limits in $\mathbb{R}^n$.
- Pointwise convergence of sequences of functions.
- Uniform convergence: definition and intuition.
- Why uniform convergence matters: exchange of limits; preservation of continuity; interchange of limit and integral; interchange of differentiation and integration (stated, not proved).

3.3 Continuity (Rigorous Intuition)

- ε - δ definition (light treatment; primarily conceptual).
- Continuity on sets versus at a point.
- Continuity of demand, supply, profit, and value functions.
- Discontinuities: types and economic meaning.
- When discontinuities create problems for optimization or equilibrium.

3.4 Compactness

- Definition of compact set (in $\mathbb{R}^n$, via closed and bounded for this level).
- Heine–Borel property (statement).
- Why compactness matters: existence of maxima and minima for continuous functions.

- Weierstrass Extreme Value Theorem (statement).
- Maximum and minimum of continuous functions on compact domains.

3.5 Product Spaces

- Cartesian products $\mathbb{R}^n$.
- Notation for vectors and product domains.
- Constraints and choice sets as subsets of product spaces.
- Separability across dimensions (conceptual).

3.6 Separation Theorem

- Convex sets in $\mathbb{R}^n$ (definition sufficient for separation; detailed convexity on Day 4).
- Hyperplane separation of disjoint convex sets (picture-based intuition).
- Supporting hyperplanes.
- Application to welfare economics: Pareto efficiency and competitive equilibrium.
- First and Second Welfare Theorems (high-level).

3.7 Fixed Point Theorems

- Fixed point: definition and economic motivation (equilibrium as fixed point).
- At least one fixed point theorem (e.g., Brouwer or contraction mapping).
- Role in proving existence of equilibrium.
- Contraction mapping as a special case (if covered).

4 Day 4 — Optimization

Approximate duration: 3 hours.

4.1 Convex Sets and Convex Functions

- Definition and geometry of convex sets.
- Convex and concave functions.
- Upper and lower contour sets (level sets).
- Why convexity or concavity guarantees global optima from first-order conditions.

4.2 Quasiconcavity and Quasiconvexity

- Definition of quasiconcave and quasiconvex functions via contour sets.
- Relationship between concavity/convexity and quasiconcavity/quasiconvexity.
- Why quasiconcavity is the standard regularity condition in utility maximization.
- When first-order conditions are sufficient for a global optimum under quasiconcavity.
- Strict quasiconcavity and uniqueness of optima (conceptual).

- Invariance under monotone transformations.
- Distinction between quasiconcavity of the objective and convexity of the feasible set in constrained problems.

4.3 Unconstrained Optimization

- Local versus global optima.
- First-order conditions: necessary conditions for interior optima.
- Second-order conditions: curvature and nature of critical points.
- Concavity and convexity of objective functions and uniqueness of optima.
- Economic optimization problems as a class (profit, utility, cost minimization).

4.4 Constrained Optimization: Equality Constraints

- Feasible set defined by equality constraints.
- Lagrange multiplier method: setup and interpretation.
- Lagrange multipliers as shadow prices or marginal values of constraints.
- Necessary conditions for constrained optima (Lagrange theorem, stated).
- Utility maximization subject to budget constraint (as problem type).

4.5 Constrained Optimization: Inequality Constraints

- Feasible set with inequalities.
- Binding versus non-binding constraints.
- Complementary slackness (conceptual).
- Karush–Kuhn–Tucker (KKT) conditions (statement and interpretation).
- Nonnegativity constraints and corner solutions.

4.6 Envelope Theorem

- **Placement:** after unconstrained and constrained optimization (FOCs, Lagrange, KKT).
- Value function of a parameterized problem: $V(\theta) = \max_x f(x; \theta)$ subject to constraints.
- **Unconstrained envelope theorem:** $\frac{dV}{d\theta} = \frac{\partial f}{\partial \theta} \Big|_{x=x^*(\theta)}$ (no need to differentiate the optimizer $x^*(\theta)$ directly).
- **Constrained envelope theorem:** $\frac{dV}{d\theta} = \frac{\partial \mathcal{L}}{\partial \theta} \Big|_{(x^*, \lambda^*)}$ with binding multipliers capturing constraint movements.
- Interpretation: shadow prices and comparative statics from the optimized value.
- Economic examples: profit as a function of a cost shifter or output price; indirect utility as a function of income and prices; expenditure/cost functions and Shephard-type logic (conceptual).
- Contrast with naive chain rule through $x^*(\theta)$: envelope theorem drops terms that vanish at the optimum.

5 Day 5 — Dynamic Programming

Approximate duration: 3 hours.

5.1 Motivation and Framework

- Sequential decision-making over time.
- Intertemporal trade-offs.
- Finite versus infinite horizon (finite focus for bootcamp).
- Markov structure: current state summarizes relevant history.

5.2 State Variables versus Control Variables

- State: variables that evolve according to law of motion.
- Control: variables chosen each period.
- Transition equation or law of motion.
- Feasibility constraints in dynamic problems.
- Classic problem types: consumption–savings, capital accumulation, resource extraction.

5.3 Bellman Equation

- Value function: maximum attainable payoff from a given state.
- Principle of optimality (Bellman’s idea).
- Recursive formulation of dynamic problems.
- Bellman equation for finite-horizon problems.
- Terminal condition and backward solution logic.

5.4 Backward Induction

- Solving finite-horizon problems from the last period backward.
- Relationship between period-by-period optimization and Bellman recursion.
- Policy function: mapping from state to optimal control.

5.5 Value Function Iteration (Conceptual)

- Idea of iterating on the Bellman operator.
- Convergence to fixed point of Bellman equation (intuition only, no algorithm).
- Distinction between solving finite-horizon and infinite-horizon problems.

5.6 Two-Period Models as Building Blocks

- Structure of two-period consumption–savings problems.
- Euler equation (conceptual introduction if appropriate).
- Link between dynamic programming and standard intertemporal optimization.

6 Cross-Cutting Themes (All Days)

- Mathematical notation used in economics graduate courses.
- Reading and interpreting first-order conditions, second-order conditions, and comparative statics.
- Connecting mathematical objects to economic meaning (marginal, shadow price, value function).
- Confidence with standard problem templates used in fall courses.

7 Suggested Time Allocation

Day	Block 1 (~1 hr)	Block 2 (~1 hr)	Block 3 (~1 hr)
1	1.1–1.4: notation; logic; sets; functions	1.5–1.6: sup/inf; algebra; Cobb–Douglas	1.7–1.9: derivatives; Euler; multivariate; limits
2	Implicit Function Theorem; MVT	Integration; Leibniz rule; Taylor	Vectors; matrices; Jacobians
3	Open and closed sets	Convergence; continuity; product spaces	Compactness; separation; fixed points
4	Convexity; quasiconcavity	Unconstrained optimization	Lagrange; KKT; envelope theorem
5	Framework; state and control	Bellman equation	Backward induction; VFI idea